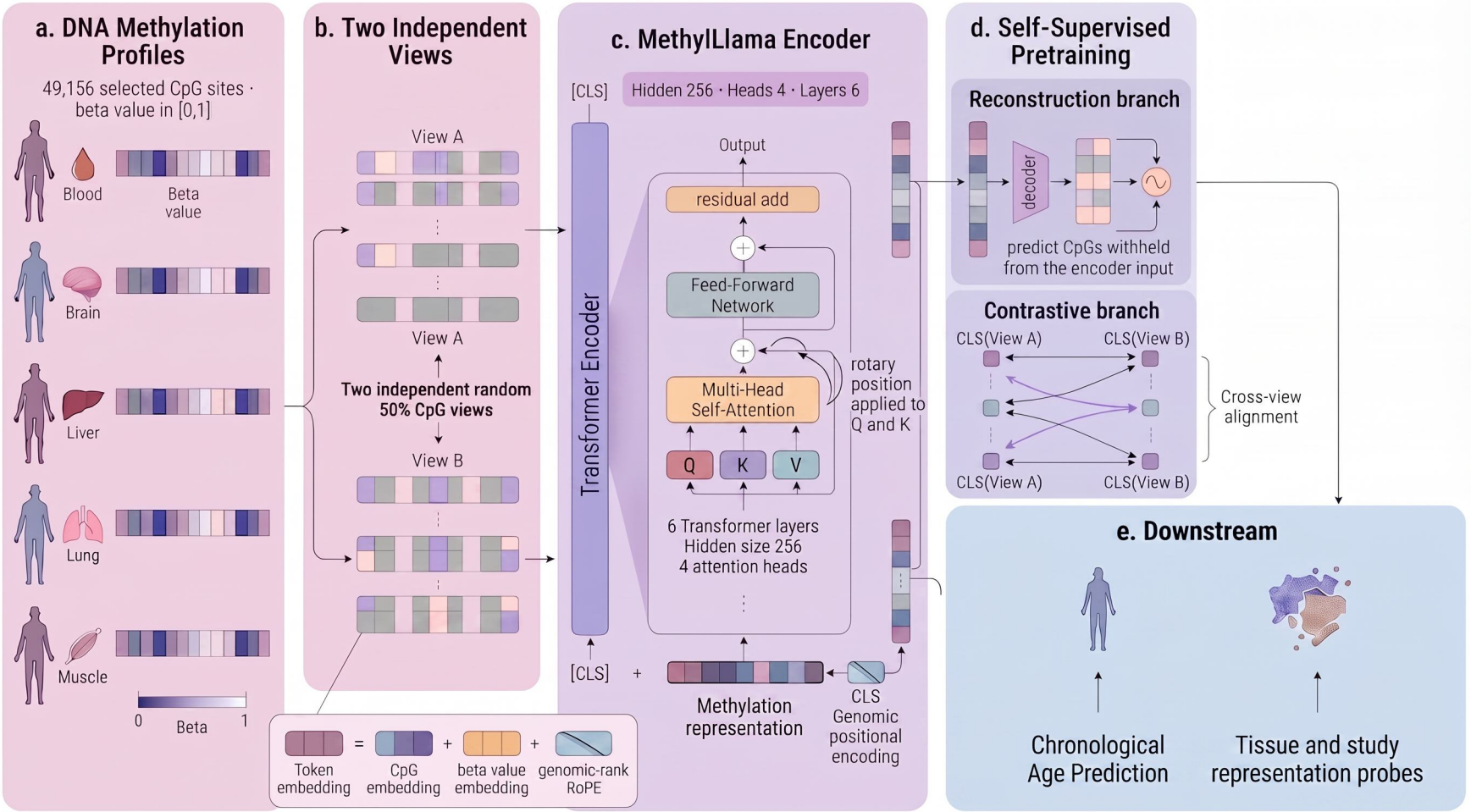

a. DNA Methylation Profiles
49,156 selected CpG sites · beta value in [0,1]
Blood
Beta value
Brain
Liver
Lung
Muscle
0
Beta
1
b. Two Independent Views
View A
View A
Two independent random 50% CpG views
View B
c. MethylLlama Encoder
[CLS]
Hidden 256 · Heads 4 · Layers 6
Transformer Encoder
Output
residual add
Feed-Forward Network
Multi-Head Self-Attention
Q
K
V
rotary position applied to Q and K
6 Transformer layers
Hidden size 256
4 attention heads
[CLS]
+
Methylation representation
CLS Genomic positional encoding
Token embedding
=
CpG embedding
+
beta value embedding
+
genomic-rank RoPE
d. Self-Supervised Pretraining
Reconstruction branch
decoder
predict CpGs withheld from the encoder input
Contrastive branch
CLS(View A)
CLS(View B)
CLS(View A)
CLS(View B)
Cross-view alignment
e. Downstream
Chronological Age Prediction
Tissue and study representation probes